CAPSTONE PROJECT

Drivers of National Economic Resilience

A panel study of 196 economies, with a five-year forecast benchmarked against the IMF

Dataset IMF World Economic Outlook, April 2026 vintage · 2,901 country-year observations · 25 variables · 2016–2030

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COURSE

Predictive Analytics

Capstone Project

FACULTY

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Problem statement

A What makes an economy resilient?

The dataset assigns every country-year a composite Economic Resilience Score (0–100), but its construction is undocumented. Which fundamentals drive it, how much can be explained, and can a country be classified as high-resilience from its fundamentals alone?

OLS · Logistic regression · Decision tree

B Where is India's output heading?

Can classical smoothing methods reproduce the nominal GDP trajectory of the fastest-growing large economy in the panel — and how does a purely statistical forecast for 2026–2030 compare with the IMF's own judgement-based staff projection?

Moving average · SES · Holt's double exponential

Why this dataset answers both

The April 2026 vintage contains realised outcomes for 2016–2025 and IMF staff projections for 2026–2030 in the same file. The historical block trains and validates every model; the projection block becomes an independent, institution-grade benchmark for Part B that no purely statistical exercise would otherwise have.

Takeaway One dataset, two complementary questions — an explanatory cross-section and a validated forecast.

Feature and target variables

TARGET VARIABLES

Model	Target	Type
OLS regression	Economic_Resilience_Score	Continuous 0–100
Logistic / tree	High_Resilience (score ≥ 70)	Binary, engineered
Time series	GDP_Nominal_USD_Bn (India)	Continuous, annual

Excluded by design

Real_Purchasing_Power_Growth
Verified to be the exact identity growth − inflation (max |residual| = 0 over 2,861 rows). Including it makes the design matrix singular.

Govt_Revenue · Import & Export volumes
Missing for 61%, 52% and 52% of rows. Imputing a majority of a column manufactures data.

GDP_Growth_Momentum_3Y
A 3-year rolling mean that contains year t — not observable in real time at the moment of forecast.

FEATURE VARIABLES (PREDICTORS)

Feature	Economic meaning	Exp.
Real_GDP_Growth_Pct	Annual real output growth	+
Inflation_CPI_Pct	Consumer price inflation	−
Govt_Gross_Debt_Pct_GDP	Sovereign leverage	−
Govt_Net_Lending_Pct_GDP	Fiscal balance	+
Current_Account_Pct_GDP	External balance	+
Unemployment_Rate_Pct	Labour-market slack	−
log_GDP_Per_Capita	Development level	+
log_Population	Economy size	?

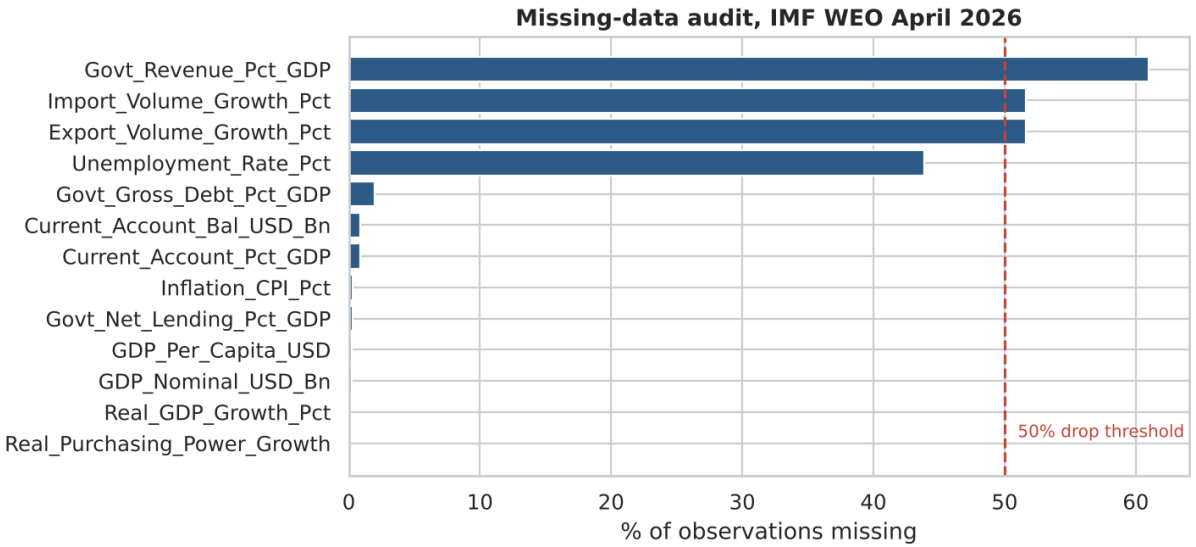
196
countries

1,890
modelling rows

2016–25
training window

Takeaway Three columns excluded on construction and missingness grounds before any model is fitted — exclusions are justified, not convenient.

Data quality and variable classification



Three columns sit beyond the 50% threshold and are dropped; unemployment (43.9%) is imputed by country median, then global median.

Classification by measurement scale

Scale	n	Examples
Nominal	3	iso_code, country_name, data_vintage
Binary	2	is_forecast, is_aggregate_region
Interval	6	Resilience score, current account, fiscal balance
Ratio	13	GDP levels, growth, debt, population, CPI
Date	1	scrape_date

Why scale matters

Measurement scale decides which statistic is legitimate. A mean is meaningful on ratio and interval variables but not on nominal ones; ratios ("twice as indebted") are valid only on ratio scales — not on the resilience score or the current account, which have arbitrary or sign-crossing zeros.

2,901 rows in file

1,953 actuals 2016–25

1,890 after cleaning

192 countries retained

Takeaway 97% of the usable panel survives cleaning — 1,890 rows across 192 countries, with every exclusion documented.

Central tendency and dispersion

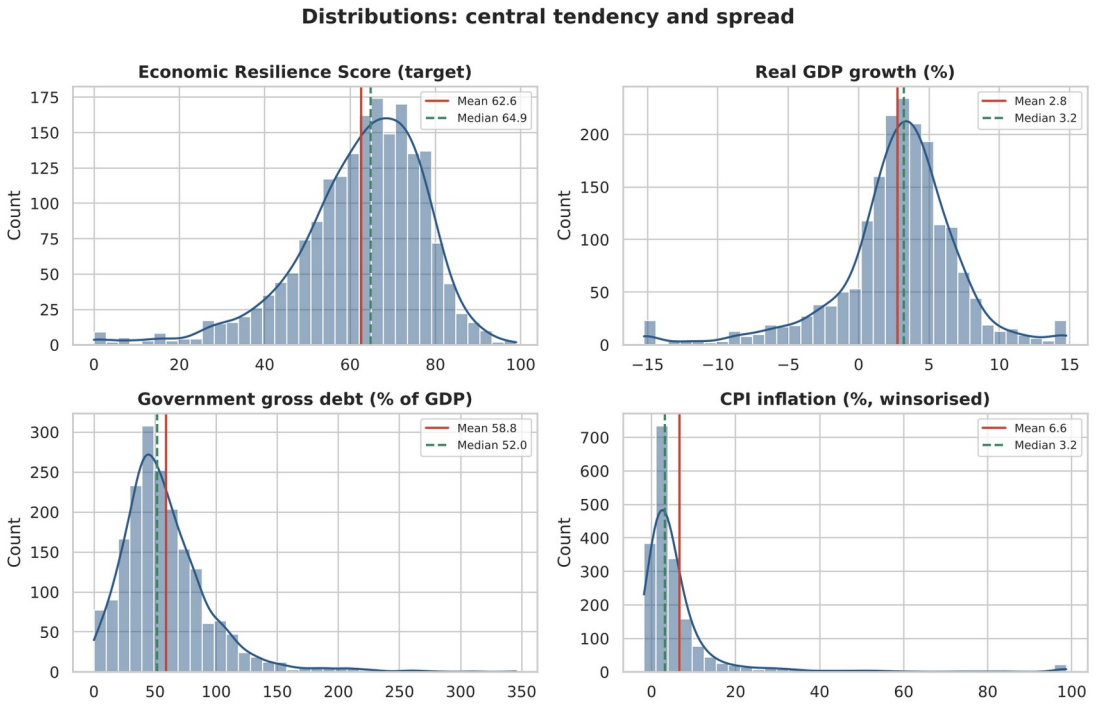
Variable	Mean	Median	Std dev	IQR	Skew	CV %
Resilience score (target)	62.6	64.9	14.9	17.9	-1.06	24
Real GDP growth %	2.8	3.2	4.5	3.9	-1.06	161
CPI inflation %	6.6	3.2	13.1	4.9	5.03	198
Govt gross debt % GDP	58.8	52.0	37.2	38.4	1.91	63
Current account % GDP	-2.0	-2.1	9.5	8.7	-0.14	-476
Unemployment %	7.3	6.2	4.8	1.7	3.82	67

Reading the table

Target is mildly left-skewed. Skew −1.06, mean 62.7 vs median 64.9 — a modest tail from crisis country-years. OLS assumes normal residuals, not a normal target, and the Q–Q plot confirms the residuals behave.

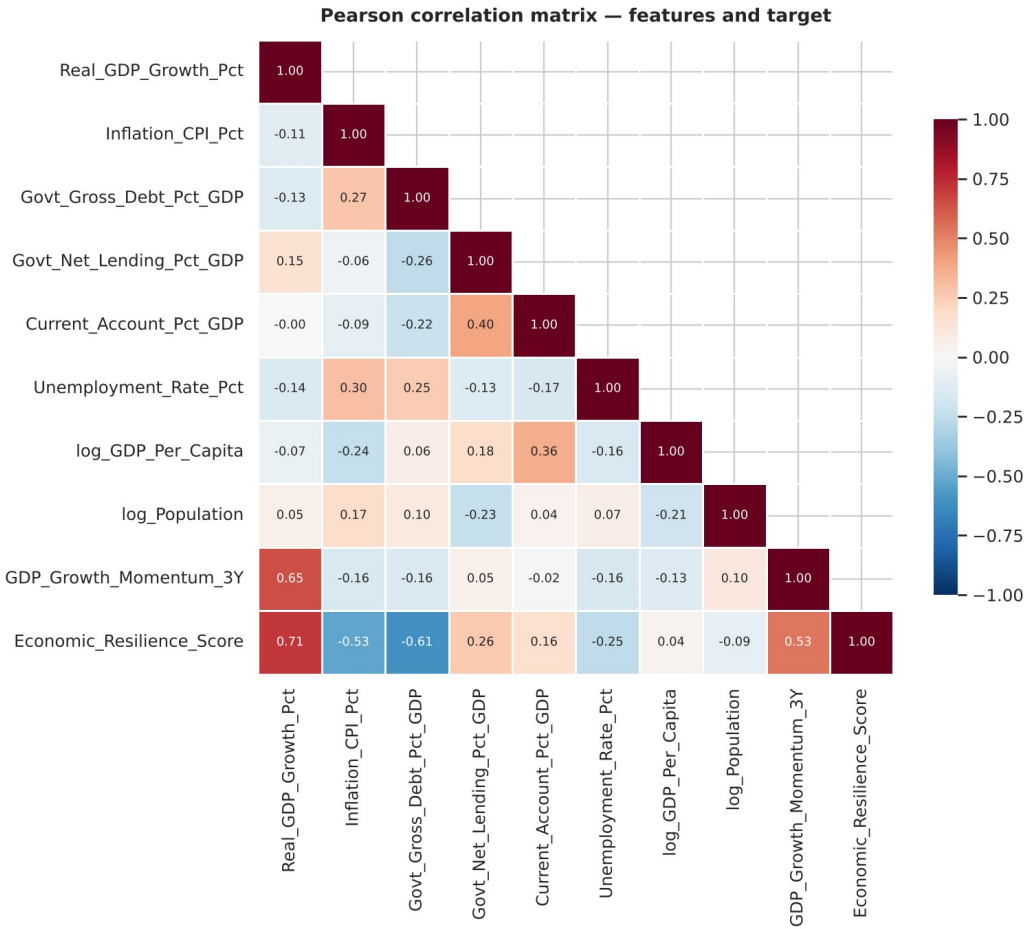
Inflation is the extreme case. Skew 5.03 and kurtosis 29.4 even after winsorising — which is exactly why winsorising was applied.

CV flags where the mean misleads. For the current account (−476%) and the fiscal balance the CV exceeds 100% in absolute value because both straddle zero.

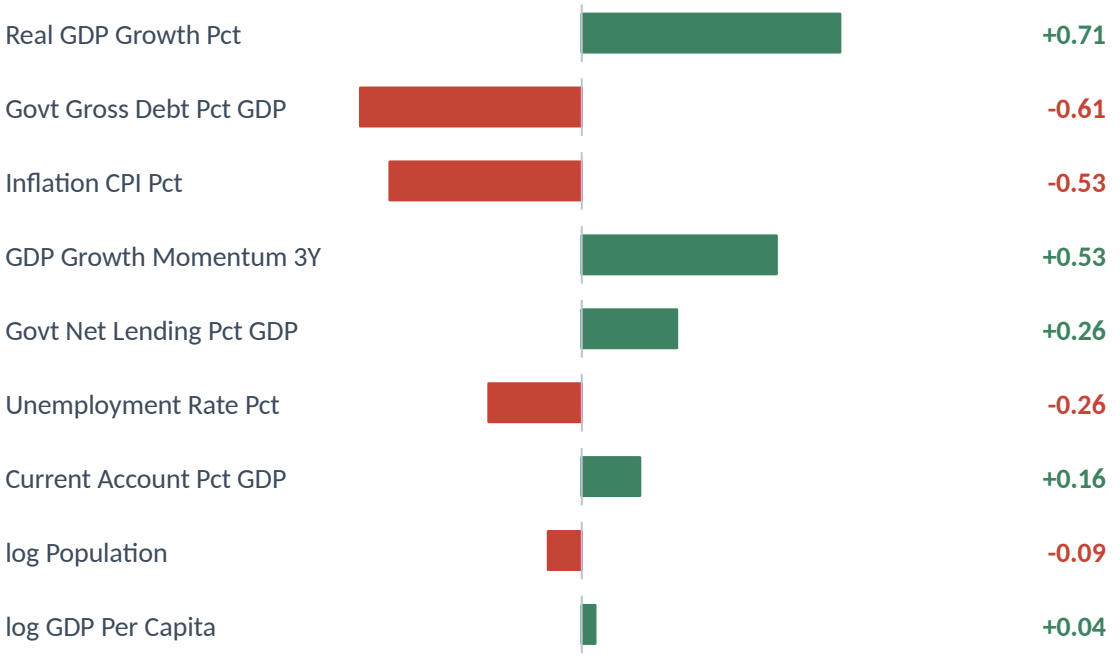


Takeaway A bounded target with a modest left tail; inflation's skew of 5.0 is what forced winsorising before any model was fitted.

Correlation analysis



Correlation with the resilience score



Pearson r, modelling sample (n = 1,890)

Being rich ≠ being resilient: log GDP per capita correlates at just +0.04.

Takeaway Growth (+0.71), debt (−0.61) and inflation (−0.53) dominate; development level is essentially uncorrelated.

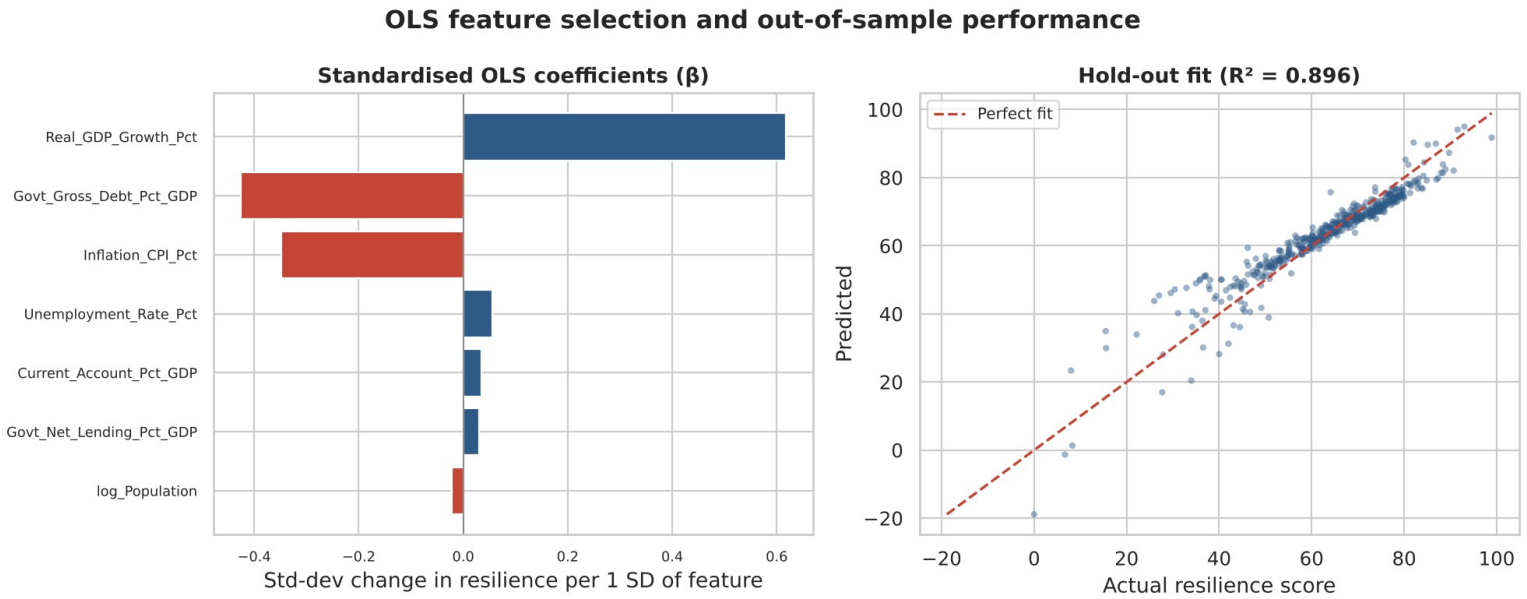
Multiple linear regression (OLS)

0.874
Adjusted R^2, training

0.896
Hold-out R^2 (25% split)

4.73
RMSE, points of 100

1,402
F-statistic, p < 0.001



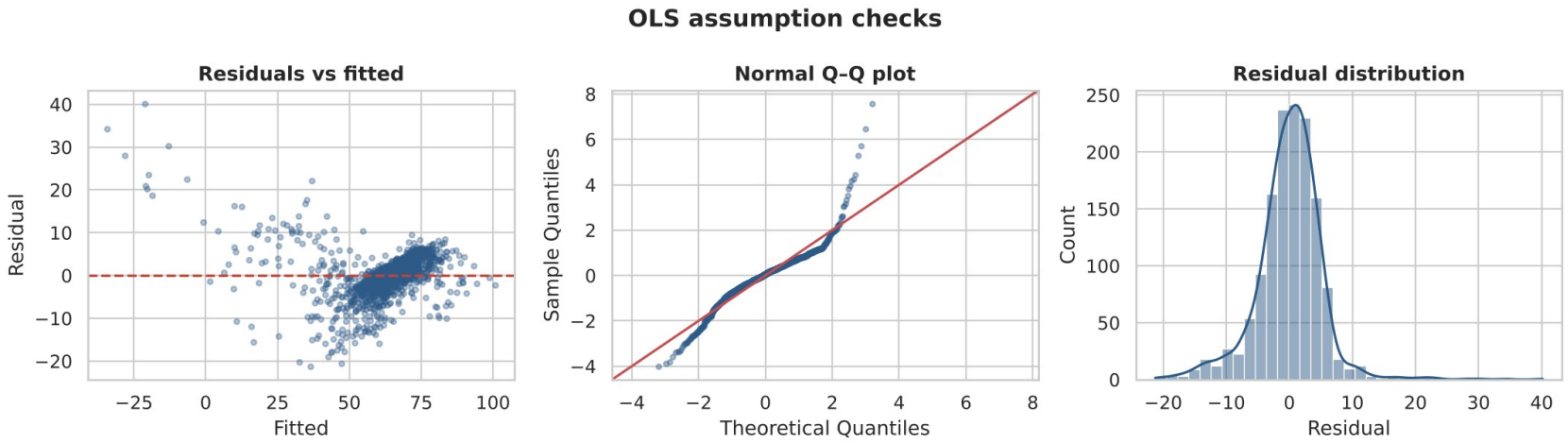
Backward elimination

Start: 8 features, all VIF < 2 — no multicollinearity problem.
Step 1: drop log_GDP_Per_Capita (p = 0.38).
Stop: all 7 remaining features significant at α = 0.05.

Hold-out R^2 exceeds training R^2 — no overfitting. Every retained coefficient carries its predicted sign.

Takeaway Seven fundamentals explain 87% of the resilience score; growth (β = +0.62), debt (−0.43) and inflation (−0.35) carry nearly all of it.

Are the OLS assumptions satisfied?



Linearity

Satisfied

Residuals vs fitted show no systematic curvature — the linear specification is not missing an obvious non-linear term.

Independence

DW = 1.99

Durbin–Watson sits almost exactly at 2, so there is no residual autocorrelation in the pooled sample.

No multicollinearity

All VIF < 2

Every variance inflation factor is far below the threshold of 5, so coefficients are separately identified.

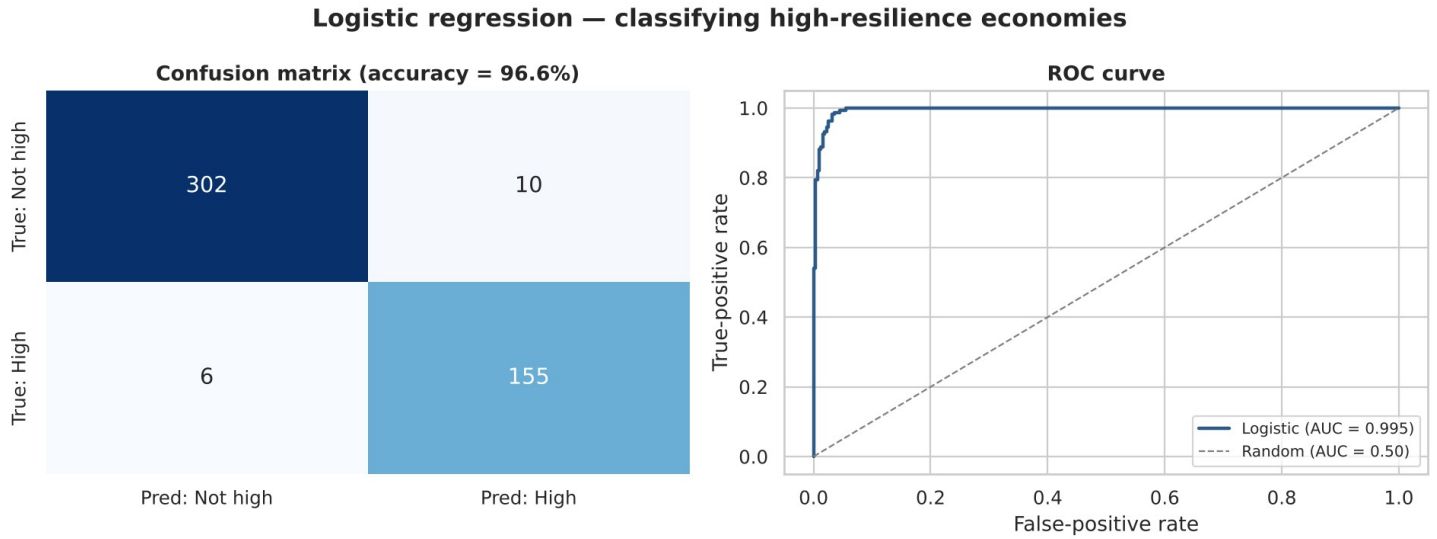
Normality of residuals

Rejected

Jarque–Bera rejects (kurtosis 9.57): residuals are symmetric but heavy-tailed, from crisis country-years.

Takeaway Three of four assumptions hold outright; non-normal residuals are harmless at n = 1,417 because the CLT still delivers valid inference.

Logistic regression — classifying high-resilience economies



96.6%
Accuracy

0.995
ROC-AUC

34%
Base rate (class 1)

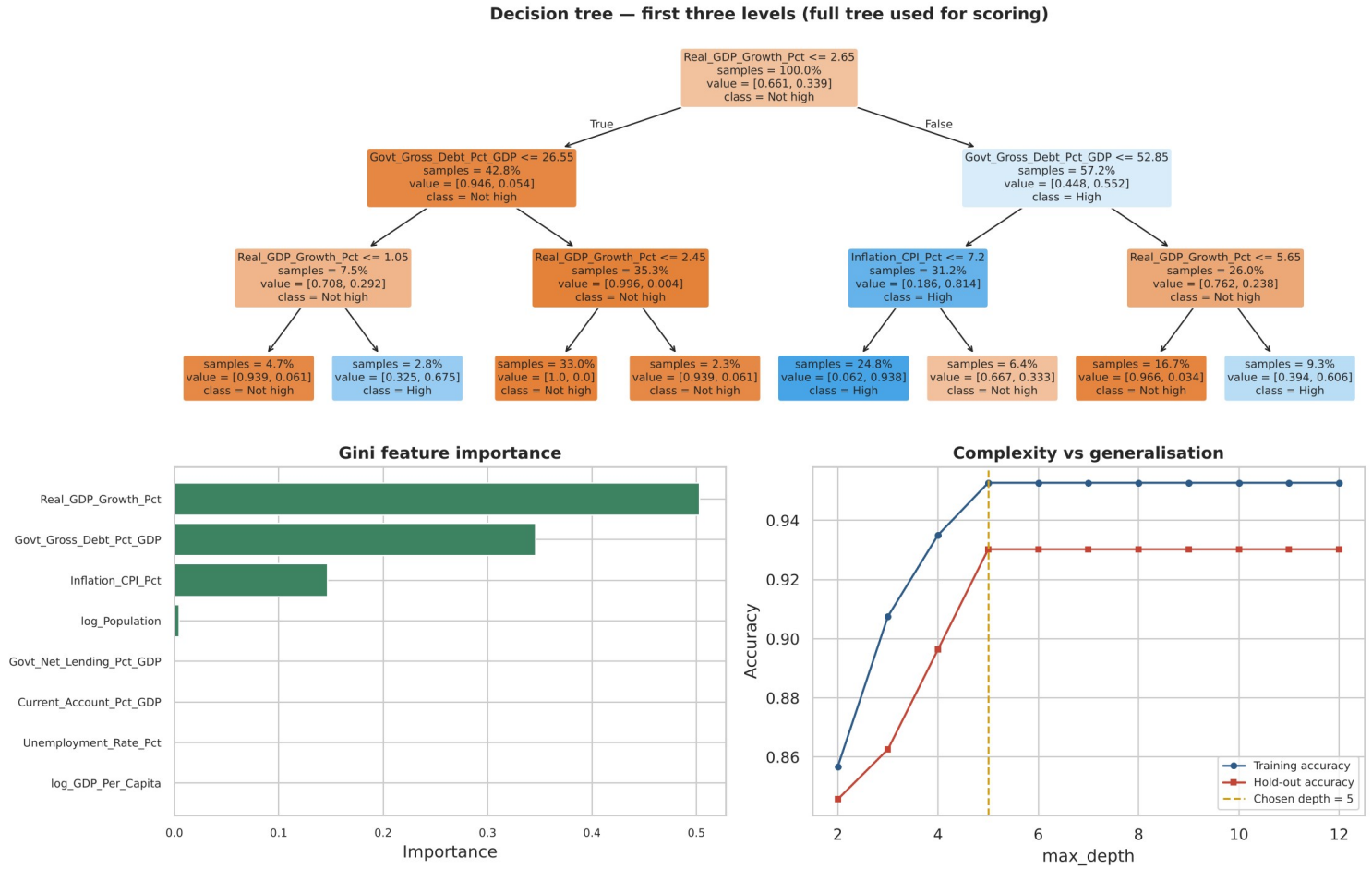
Odds ratios, per 1 SD of the feature

Feature	Odds ratio	Effect
Growth	493	raises
Debt	0.008	cuts
Inflation	0.009	cuts
Fiscal bal.	1.057	raises
Curr. acct	1.016	raises
Unemploy.	0.959	cuts

Reading the odds. One extra standard deviation of growth multiplies the odds of high-resilience status by a very large factor; a standard deviation of debt or inflation cuts them by roughly two orders of magnitude (≈0.008 and ≈0.009).

Takeaway High-resilience status is near-perfectly separable from fundamentals: AUC 0.995 against a 34% base rate.

Decision tree classifier



93.0%

Hold-out accuracy

0.970

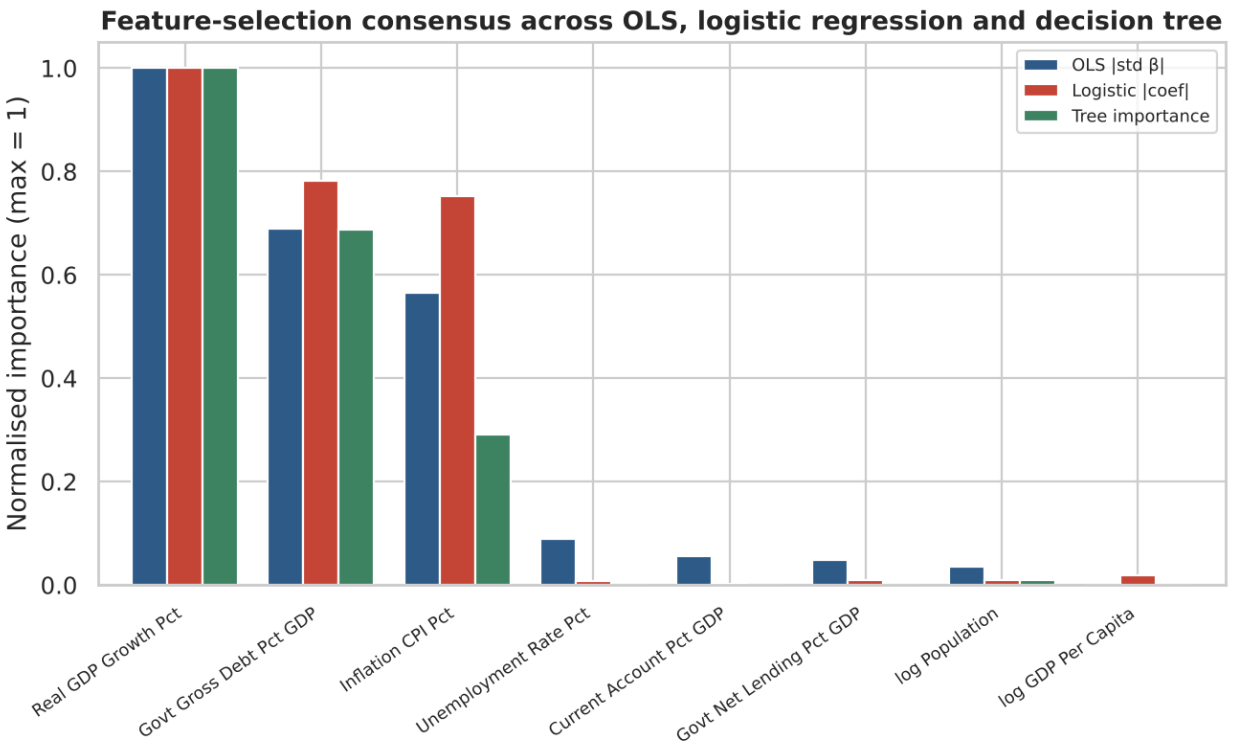
ROC-AUC

Pruned to depth 5

Hold-out accuracy plateaus beyond depth 5 while training accuracy keeps climbing — the textbook overfitting signature, and the reason for pruning.

Takeaway The tree splits first on growth, then on debt, then on inflation — three features carry 99.5% of all splitting power.

Feature-selection consensus across three model families



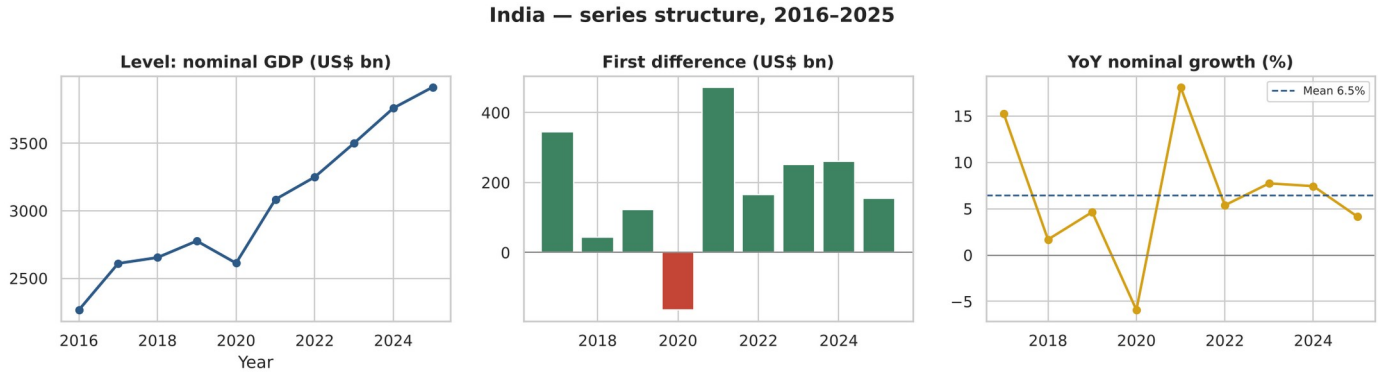
Three methods, one answer

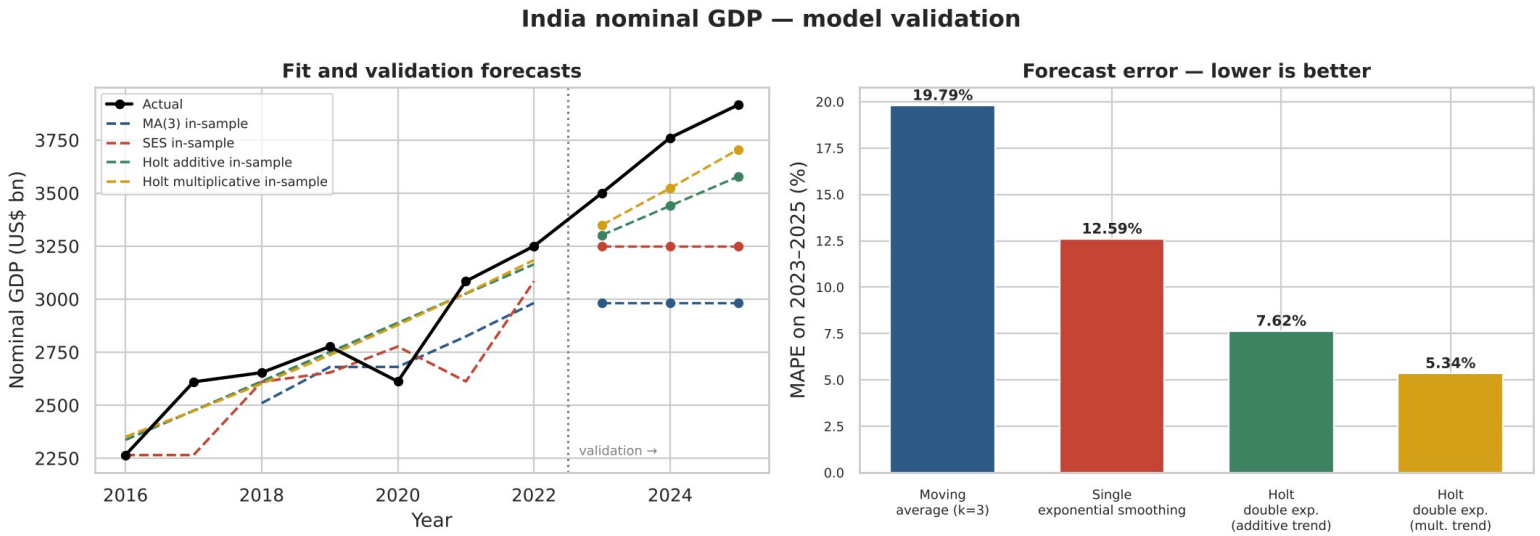
1	Real GDP growth	1.00
2	Govt gross debt	0.72
3	CPI inflation	0.54
4	Everything else	< 0.04

Normalised consensus score — the mean of standardised OLS |β|, logistic | coefficient| and Gini importance, each scaled so the leader equals 1.00.

Takeaway A parametric linear model, a parametric non-linear model and a non-parametric model independently rank the same three features 1-2-3.

India's nominal GDP — structure and method validation





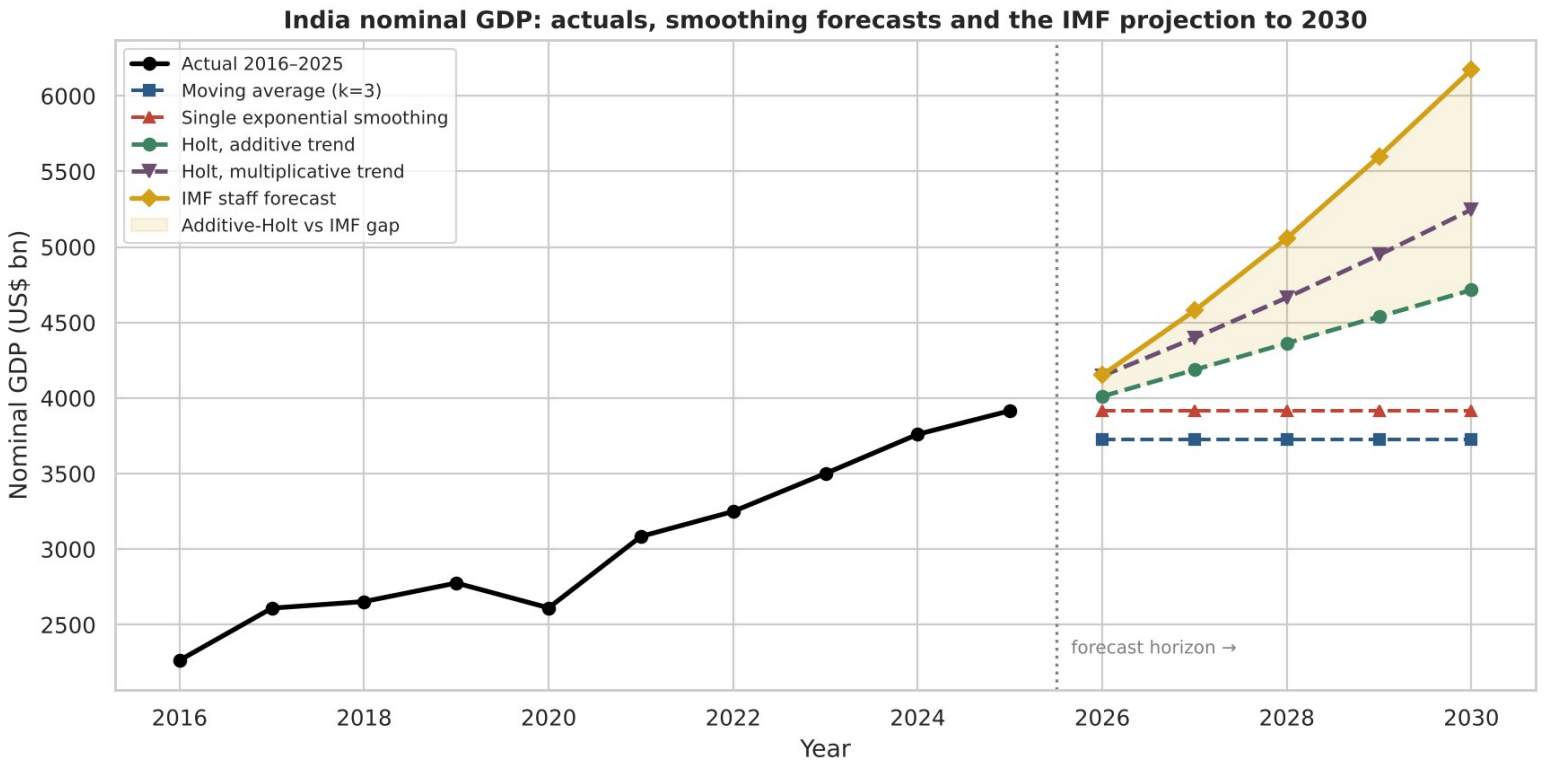
Validation, 2023–2025

Method	MAPE
Moving average (k=3)	19.79%
Single exp. smoothing	12.59%
Holt — additive trend	7.62%
Holt — mult. trend	**5.34%**

Level-only methods are structurally biased against a trending series: their forecast function is a flat line.

Takeaway Adding a trend component cuts forecast error by 62%; a multiplicative trend cuts it by 73%.

Forecast to 2030 versus the IMF staff projection



2030 outcomes, US$ bn

Forecast	2030	vs IMF
IMF staff	**6,173**	—
Holt mult. trend	5,248	-15.0%
Holt additive	4,715	-23.6%

Decomposing the gap

Functional form. A constant-increment model cannot track a compounding path — worth ~9 pp of the gap.

Assumed acceleration.
The IMF's implied 9.5% nominal CAGR exceeds India's realised 6.3% over 2016–2025.

6.3% Realised 2016–25

6.0% Holt forecast

9.5% IMF 2025–30

Implied nominal CAGR

Takeaway The IMF's India projection is not trend persistence — it embeds a 3.3 pp acceleration no univariate model can see.

Conclusions

1

Three levers explain resilience

Growth, debt and inflation rank 1-2-3 by standardised β, by log-odds and by Gini importance — 99.5% of the tree's splitting power.

2

Being rich ≠ being resilient

GDP per capita is the one feature backward elimination removes ($p = 0.38$). Resilience is about policy outcomes, not development level.

3

Near-perfect separability

High-resilience status classifies at AUC 0.995. The linear model beating the tree signals a close-to-linear boundary.

4

Structure dictates method

Against a trending series, level-only methods are structurally biased: MAPE 19.8% versus 5.3% once trend is modelled.

5

The gap measures judgement

The IMF's 9.5% implied CAGR exceeds India's realised 6.3%. The distance from the naive baseline quantifies institutional judgement.

Limitations

Panel observations are not independent, so standard errors are optimistic — fixed effects are the next step. Unemployment is imputed for 44% of rows. The resilience index is a vendor composite. The forecast rests on ten annual observations for one country.

Takeaway Three fundamentals explain resilience across every method tried; method structure, not parameter tuning, drives forecast accuracy.

Reproducibility and repository

Everything in this deck is reproducible from a single notebook

IMF_WEO_Capstone_Analysis.ipynb

44 cells, fully executed, all outputs retained

figures/ — 11 charts

Every figure in this deck, generated by the notebook

imf_weo_april_2026.csv

Source data, 2,901 rows × 25 columns

ACTIVE PROFILE LINK

Repository
github.com/swayd03/
imf-weo-capstone

GitHub profile
github.com/swayd03

Submitted in partial fulfilment of the Predictive Analytics capstone · **Swadha Singh** · 246101554 · Prof. Dr. Varsha P S

Methods: OLS · Logistic Regression · Decision Tree · Moving Average · Single & Double Exponential Smoothing